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APPENDIX A. METHODOLOGY

1. Introduction

In this appendix, we outline the survey methodology used for the *2004 Survey of Household Energy Use* (SHEU), in particular the activities undertaken and the issues addressed by the methodologists of the Social Survey Methods Division (SSMD) and the Business Survey Methods Division (BSMD) of Statistics Canada.

This survey was conducted in 2004 by Statistics Canada, and is therefore referred to in this methodology section as the *2004 Survey of Household Energy Use* (SHEU-2004). However, the reference period for this survey is the calendar year 2003 (that is, all data presented are for 2003 calendar year). Therefore, all other sections of this report and the summary report refer to the survey as the *2003 Survey of Household Energy Use* (SHEU-2003).

This outline has been adapted from a document prepared by Statistics Canada.

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2. Pre-collection

The pre-collection period of a survey begins with the initial designing of the project and ends once data collection has started. It is a critical period, for it establishes the foundation for the entire project.

2.1 Survey objectives

The 2004 SHEU builds on the surveys of the same name that were conducted in February 1993 and February 1998. They served to gather data on the energy use characteristics of private dwellings in Canada and on household use of energy resources. The 2004 data will be used to assess the effectiveness of existing energy efficiency programs and to develop new ones.

The target population for the 2004 SHEU was comprised of all dwellings that were occupied as primary residences in the ten Canadian provinces[1] and that fit into one of the following categories: single detached, semi-detached, row, mobile, duplex or dwelling in a building with no more than four storeys (the last two categories were not covered in the 1998 edition of the

[1] The territories are not included in the target population.

SHEU). Specifically excluded from the survey's coverage were dwellings not mentioned above, dwellings located in a First Nation community or on a military base, businesses, institutions, demolished dwellings, dwellings under construction, seasonal or secondary residences, and dwellings occupied by individuals who work full-time within the Canadian Armed Forces.

As was the case in 1998, the survey sought to obtain energy consumption data directly from suppliers of electricity, natural gas, fuel oil and propane. Signed authorizations from households were required before we could contact the energy suppliers for this data. In the case of tenants, we also had to contact the non-occupant owner for certain pieces of information and to obtain authorization to contact the energy suppliers.

The sample of dwellings for the survey was designed to produce reliable estimates for each of the five large Canadian regions (Atlantic provinces, Quebec, Ontario, Prairie provinces, British Columbia) as well as national estimates for certain aggregate variables, such as groupings by type of dwelling or year built, and by rural or urban location.

The survey was conducted on behalf of Natural Resources Canada. Respondents, including non-occupant owners, where applicable, were informed of a data-sharing agreement under which Natural Resources Canada would have access to a complete microdata file.

2.2 Survey frame

The two main sources of data first envisaged for the SHEU survey frame were the *Labour Force Survey* (LFS) and the *Canadian Community Health Survey* (CCHS). One disadvantage of the LFS was that we had to select whole rotation groups if we wanted to avoid unduly complex variance estimation. One significant advantage of the CCHS over the LFS was that the CCHS allowed for obtaining reduced design effects, meaning that the required sample could be reduced, along with the related collection costs. The CCHS was therefore selected as the survey frame for the SHEU.

More specifically, cycle 2.1 of the CCHS was selected because its auxiliary information was current. The CCHS area frame was targeted because its coverage of the territory covered by the SHEU, i.e., the 10 Canadian provinces, was excellent. The respondents for the collection months from January to August 2003 were identified as being available for the SHEU sample.

2.3 Sample design

What follows is an overview of the sample design used for the *2004 Survey of Household Energy Use*.

The SHEU sample was composed of respondents from the area frame of the *Canadian Community Health Survey* (CCHS) who were assigned to collection months ranging from January to August 2003. They were interviewed for the CCHS between January and October 2003 (the number of interviews done in September and October is low because only households previously flagged as unresolved were interviewed during these months). The auxiliary information available for SHEU purposes – type of dwelling, telephone number and occupant type (tenant or owner) – was therefore relatively recent when we established the sample in January 2004; the information had been gathered during the previous three to twelve months and was six months old on average.

The SHEU sample is a stratified simple random subsample of CCHS respondents. The SHEU strata definitions are based on the provincial health regions, with some slight modifications to avoid overlap between strata of the *Labour Force Survey* (LFS) and those of the health regions (LFS strata not assigned to a single health region were assigned to one). The sample allocation to SHEU strata was proportional to population estimates. This approach to sample design, pseudo-proportional to the weight, has the advantage of reducing design effects, resulting in estimates of better quality for given sample sizes.

The sample size required to meet the analysis needs summarized in Section 2.1 – keeping in mind the proposed survey design and the expected response rate – was 6,433 dwellings. Tables 1 to 5 provide details on the distribution of these dwellings by regional office, province, type of dwelling, CCHS participation and type of population centre.

Table 1
Sample distribution by regional office

Regional office	Frequency	Percentage
Halifax	1,089	16.93%
Montreal	1,398	21.73%
Toronto	1,424	22.14%
Edmonton	1,358	21.11%
Vancouver	1,164	18.09%

Table 2
Sample distribution by province

Province	Frequency	Percentage
Newfoundland and Labrador	252	3.92%
Prince Edward Island	65	1.01%
Nova Scotia	429	6.67%
New Brunswick	343	5.33%
Quebec	1,398	21.73%
Ontario	1,424	22.14%
Manitoba	297	4.62%
Saskatchewan	259	4.03%
Alberta	802	12.47%
British Columbia	1,164	18.09%

Table 3
Sample distribution by type of dwelling

Type of dwelling	Frequency	Percentage
Single detached house	4,125	64.12%
Semi-detached house	317	4.93%
Double/row house	353	5.49%
Duplex	225	3.50%
Apartment in a building with fewer than five storeys	1,231	19.14%
Mobile home	182	2.83%

Table 4
Sample distribution by participation in the Canadian Community Health Survey

Participant	Frequency	Percentage
Selected	1,804	28.04%
Not selected	4,629	71.96%

Table 5
Sample distribution by type of population centre

Type of population centre	Frequency	Percentage
Rural	1,234	19.18%
Urban	5,144	79.96%
Remote	55	0.85%

2.4 Interview method

Given the nature of the survey, interviews had to be conducted in person at the respondents' dwellings, primarily to assist them in answering the questions on the characteristics of their dwelling and to obtain their authorization in writing for us to contact their energy suppliers. However, the regional offices calculated that collecting data from certain respondents would be very costly if they lived in a remote area or if there were not enough interviewers in the region. Statistics Canada's head office examined the requests from the regional offices and in most cases authorized them to conduct the interviews by telephone. The code "authorization for a telephone interview" was given to 138 households, as shown in Table 6. No information was available during or after the survey for determining the number of interviews actually conducted by telephone. In principle, telephone interviews were authorized only for these 138 dwellings, or 2.1 percent of the initial sample of 6,433. In comparison, approximately 3 percent of the 1998 SHEU interviews were done by telephone.

Table 6
Distribution of dwellings for which a telephone interview was authorized

Province	**Number of dwellings for which a telephone interview was authorized**	**Initial sample**	**Proportion of dwellings for which a telephone interview was authorized**
Newfoundland and Labrador	20	252	7.9%
Prince Edward Island	0	65	-
Nova Scotia	0	429	-
New Brunswick	0	343	-
Quebec	14	1,398	1.0%
Ontario	50	1,424	3.5%
Manitoba	0	297	-
Saskatchewan	33	259	12.7%
Alberta	18	802	2.2%
British Columbia	3	1,164	0.3%
TOTAL	**138**	**6,433**	**2.1%**

2.5 Questionnaire

The 2004 SHEU questionnaire had two separate parts. The first part, the more detailed of the two, was to be completed by the occupant of the dwelling selected from the sample. The shorter second part was needed for rented dwellings and condominiums. It was to be completed by the non-occupant owner in the first case and the condominium manager[2] in the second case, since tenants and condominium occupants did not know, or were unlikely to know, the information it was designed to gather.

The questionnaire was administered by the interviewer using a computer-assisted interview system that provided a set of built-in checks, such as allowable value ranges, to ensure a certain level of quality in the data.

2.6 Letter of introduction to the survey

A letter of introduction summarizing the purpose of the survey was sent out by the regional offices of Statistics Canada to the households selected from the SHEU sample shortly before the data were to be collected. This letter was designed as an incentive to participate in the survey and as an act of courtesy toward the CCHS respondents being sampled again for the SHEU. The address used came from the SHEU frame and matched the address of residence at the time of the CCHS.

3. Data collection

Data collection includes the work of obtaining information from individuals, households and, for the SHEU, third parties such as non-occupant owners and suppliers of energy resources. It is a costly phase, yet it is extremely important as the quality of the final product depends on it.

The first phase of data collection for the SHEU involved interviewing the occupants of the sampled dwellings and, if applicable, the non-occupant owners. Data were collected from March to June 2004 using computer-assisted personal interviews. A small proportion of interviews was done by telephone, as explained in Section 2.4.

It should be noted that the first phase of the data collection for the 1998 survey was done entirely in March, which could have influenced how the survey questions were answered. For example, the heating and the air conditioning of a dwelling probably change between March and June, and it is possible that the respondent's perception of this, despite the reference to the "heating season" and "summer season" in the survey questionnaire, was influenced by the date of the interview and its separation from the periods referred to in the questions.

The second phase of data collection was coordinated by Statistics Canada's head office. It involved gathering energy consumption data from suppliers in cases where the occupant or non-occupant owner had given his or her consent.

Respondents who took up residence in their dwelling during 2004 were not interviewed but were flagged as non-respondents when the weighting was done. The main reason for this was the fact that they would not have any energy consumption data for the 2003 reference year. Respondents

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[2] For the sake of brevity, in the remainder of this report "non-occupant owner" means the non-occupant owner or the condominium manager, as the case may be.

who took up residence in their dwelling in 2003 were interviewed. As expected, their rates of imputation for the energy use variables are higher than the rates for the rest of the sample (except for propane).

3.1 Response rate

In order to define the response rate for the first collection phase, the SHEU sample was first divided into three mutually exclusive groups (see Table 7).

Table 7
Classification of the sample

Group	Definition	Sample
Respondent	Dwelling for which we obtained sufficient data and an authorization for data sharing.	4,551
Non-respondent	Dwelling for which we did not obtain sufficient data or an authorization for data sharing.	1,573
Unit out of scope	A dwelling whose type is incompatible with the survey: dwelling located on a military base or in a business or institution; dwelling vacant, demolished or under construction; seasonal or secondary residence; or dwelling located in a First Nation community or occupied by individuals who work full-time within the Canadian Armed Forces.	309
Total		**6,433**

We included the criteria of authorization for data sharing in the definition of “respondent” since the rate of authorization was relatively high (98 percent for occupants, 94 percent for non-occupant owners) and so we would have to produce only one file of “sharing” respondents for the survey.

Table 8 shows the three types of response rates calculated.

Table 8
Definition of response rates

Rate 1	Raw response rate	= respondents/sample
Rate 2	Intermediate response rate, where out-of-scope units are omitted from the denominator	= respondents/(sample – out-of-scope units)
Rate 3	Operational response rate	= (respondents + out-of-scope units)/sample

It should be noted that these definitions do not accurately reflect the work performed in the field. In fact, interviews classified as “complete” by regional office staff may have been reclassified as “non-respondent” or “out of scope” based on the above criteria.

By definition, these three response rates are necessarily in ascending order, from Rate 1 to Rate 3.

Table 9 summarizes the response rates of the first collection phase and the rate of out-of-scope dwellings, the rate of refusal and the rate of unoccupied dwellings.

Table 9
Response rate and other unweighted rates of the SHEU

Raw response (Rate 1)	70.7%
Intermediate response (Rate 2)	74.3%
Operational response (Rate 3)	75.5%
Rate of out-of-scope dwellings	4.8%
Rate of refusal	9.7%
Rate of unoccupied dwellings	2.5%

3.2 Data sharing

In light of the data-sharing agreement between Statistics Canada and Natural Resources Canada, respondents were asked for their authorization to share the gathered information. The question was worded as follows:

> Statistics Canada has entered into an agreement with the Office of Energy Efficiency of Natural Resources Canada (NRCan) to share information from this survey.
>
> NRCan will not be given your name or other identifiers. The information shared with them will contain sufficient geographic detail to allow them to analyse the data for small areas. NRCan has agreed to keep all the information provided confidential and to use it only for statistical purposes.
>
> Do you agree to allow Statistics Canada to share your information with NRCan?

An answer in the affirmative to that question authorized Statistics Canada to share the information gathered with Natural Resources Canada. Table 10 summarizes the rates of authorization for data sharing.

Table 10
Rates of authorization for data sharing

Authorization from occupant of dwelling	98.0%
Authorization from non-occupant owner	94.0%
Combined rate	93.0%

3.2 Rate of consent to contact suppliers of energy resources

Energy consumption data could be obtained in two ways: by consulting the suppliers of electricity, natural gas, fuel oil and propane, with the signed agreement of the respondent (occupant or non-occupant owner), or by entering data from bills or statements of account that the respondent provided to the interviewer for as long as was required to enter the data. The first method was encouraged in order to obtain more accurate data and to reduce the time spent with the respondent.

Table 11 summarizes the rates of consent to contact the suppliers of energy resources.

Table 11
Rates of consent to contact suppliers of energy resources

	Consent to contact all applicable suppliers	No consent given for one or more suppliers	Total	Rate of consent
Occupant	3,814	526	4,340	87.9%
Non-occupant owner	331	54	385	86.0%
Occupant and non-occupant owner	3,944	569	4,513	87.4%
Occupant and non-occupant owner:				
Electricity	4,021	477	4,498	89.4%
Natural gas	1,928	256	2,184	88.3%
Fuel oil	540	123	663	81.4%
Propane	77	21	98	78.6%

It should be noted that the above rates are based on the answers to questions CESk_Q01 and LCSk_Q01, where k = 1 (electricity), 2 (natural gas), 3 (fuel oil) or 4 (propane). Some dwellings using a given energy source may not have been asked for the corresponding consent, primarily because of conceptual errors in the design of the collection instrument itself. Accordingly, the rates shown are the rates of consent for the dwellings that responded to the request for consent and not for the dwellings that use the energy source in question.

In addition, once consent was given to contact a supplier, various factors may have prevented using the data. For example, Statistics Canada's regional offices did not receive some consent forms from the account holders,[3] and others were received but not signed (0.8 percent of the forms received: electricity, 0.7 percent; natural gas, 0.6 percent; fuel oil, 1.9 percent; propane, 0.0 percent). Lastly, we estimate that 15 percent of the consumption data requests made to the suppliers were unanswered (electricity, 9 percent; natural gas, 19 percent; fuel oil, 40 percent; propane, 36 percent).

We estimate that for the dwellings using natural gas, only 60 percent of the consumption data was obtained from the supplier. This rate is apparently higher for electricity and lower for fuel oil and propane. Moreover, the rates of data received from the suppliers – amongst dwellings that use a given energy source, before verification and imputation – follow suit, as can be seen in Table 12.

Table 12
Rates of data received from suppliers, before verification and imputation, by dwellings using the specified energy source

Energy source	Rate
Electricity	76%
Natural gas	64%
Fuel oil	48%
Propane	41%

[3] The probable causes included consent questions improperly marked "yes" and consent forms left at the home for signing by someone absent at the time of the interview and not returned to Statistics Canada.

4. Post-collection

The post-collection period includes all activities undertaken once the data have been collected. The ultimate aim of these activities is to produce a microdata file and documentation that analysts can use to extract the information they need for their work. To that end, many sectors within Statistics Canada contribute in a variety of ways. This is often the most complex phase of a survey, as every action can have an impact on the final quality of the data.

4.1 Processing the data

Processing the data involves a variety of activities with different purposes, such as:

- classifying the sample and identifying the respondents
- formatting the data
- cleaning up paths
- encoding some variables
- establishing the verification rules
- creating derived variables
- imputing

SSMD methodologists classified the sample and the respondents as described in Section 3.1. They also revised some rules for cleaning up the paths, suggested some derived variables and helped the BSMD methodologists verify and impute the data.

o Quality of the data

Overall, the data gathered for the SHEU are of good quality. The reports of the interviewers and the internal consistency of the data attest to this. However, some aspects of the quality of the data and some reservations emerging during the collection and processing phases are worthy of note, as described in the following paragraphs. The reports from the various sectors of the office may reveal other limitations or difficulties inherent to the survey data.

o Modification of type of dwelling between CCHS and SHEU

For 8 percent of the sample of 4,551 SHEU respondents, the type of dwelling changed after the CCHS (January to October 2003) and before the SHEU (March to June 2004). A total of 362 dwellings underwent legitimate transformations, resulting in a change in type of dwelling, or were the subject of variable responses depending on the interviewer or respondent. In particular, it seems that “duplex” and “semi-detached” types of dwellings may have been switched in some cases.

o Person answering the survey questions

When we contacted the household to make an appointment for the interview, we asked that the person who paid the utility bills be there at the time of the interview in order to sign the forms authorizing us to contact suppliers. During the 1998 survey, in comparison, it seems that the appointment for the in-home interview tended toward interviewing the person with the best knowledge of the characteristics of the dwelling, including the appliances and their use, the heating system, the windows and the dwelling area. It is therefore possible that the 2004 approach resulted in interviewing individuals less familiar with some of the characteristics measured by the survey. It is probable that the effect on the quality of the answers is relatively marginal; the rate of “don’t know” answers could be an indicator of this.

- **Non-occupant owner, duplex or apartment**

Preliminary analysis revealed that the quality of the data obtained from occupants is better than the quality of the data obtained from non-occupant owners, particularly regarding duplex dwellings and apartments located in buildings with fewer than five storeys.

- **Heated area**

According to the report “Regional Operations Branch – Interviewers’ Debriefing Summary,” the dwelling area and the building area were sometimes difficult to determine, especially for tenants. The heated portion of the basement was apparently particularly difficult to measure. It is possible that the heated area of the dwelling was difficult to determine because of a lack in the definition used for “heated room.” In addition, the survey was held between March and June, which means that the data on the reported heated area could have been influenced by the month in which the interview was held.

- **Dwelling area and building area**

Inconsistencies have been detected between dwelling area and building area. It is possible that the error stems in part from the respondent’s confusing the terms “logement” and “immeuble” in French or “dwelling” and “building” in English. These inconsistencies have been processed by verification and imputation.

- **Year built, year moved in and age of heating system**

There are some inconsistencies between the year in which the dwelling was built, the year the household moved in and the age of the heating system.

- **Missing data**

Flag variables generated by the software used to assist in data collection were used to confirm dwelling use of any one or more of the four energy sources targeted by the survey – electricity, natural gas, fuel oil and propane. For approximately 200 dwellings, the variables needed to produce these flags were empty. In these cases, it was uncertain if the dwelling used a specific energy source. In addition, the variables used to determine if the garage or basement was heated were missing for a few dozen dwellings. Because processing all of these cases would have entailed procedures too complex for the time allotted for this work, and since the impact seemed minimal, given the relatively low number of dwellings in question, these cases were declared non-respondent, and their weight was distributed among the respondent dwellings with similar characteristics.

In addition, some questions on household appliances could be answered using the “other – please specify” category without the respondent being asked which energy sources were used for the appliances. This was the case for approximately 300 dwellings, and it is therefore possible that an energy source used only for the appliance in question was not indicated. This could result in a slight underestimation of consumption, in the case of energy sources that could not be ascertained because the questions were omitted. It was recommended that we not use the dwellings affected by this problem as donors when imputing the values of the energy consumption variables.

- **Gas fireplaces**

The survey questionnaire assumed all gas fireplaces use natural gas, while some can use propane. Survey data revealed that 100 dwellings that reported having a gas fireplace did not declare natural gas use for any other equipment type. After re-evaluation, these 100 dwellings were distributed as follows: 40 use natural gas and 60 do not (10 had already indicated they use propane, and 50 will use it from now on).

Imputation

Table 13 summarizes the rates of imputation for the SHEU variables subject to imputation.

Table 13
Rates of imputation

Variable	Description	Imputed records	Non-imputed records	Total	Rate of imputation
DFS_M01	In what year was the dwelling originally built?	558	3,993	4,551	12.26%
DFS_M02	Would you say it was built in . . . ?	31	4,520	4,551	0.68%
SOD_M02	What is the heated area of your basement in square feet/metres?	290	2,175	2,465	11.76%
SOD_M03	Would you say it is . . . ?	71	2,394	2,465	2.88%
SOG_M01	What is the size of the heated area of your indoor garage in square feet/metres?	33	32	65	50.77%
SOG_M02	Would you say it is . . . ?	14	51	65	21.54%
SOD_M06	What (excluding the basement and/or garage) is the heated area of your dwelling in square feet/metres?	563	3,988	4,551	12.37%
SOD_M07	Would you say it is . . . ?	210	4,341	4,551	4.61%
MSTOTDWE	Total dwelling area heated (square feet/metres)	646	3,905	4,551	14.19%
SBD_M01	Excluding the indoor parking, what is the heated area of your building in square feet/metres?	613	184	797	76.91%
SBD_M02	Would you say it is . . . ?	413	384	797	51.82%
MSTOTBLD	Total building area heated (square feet/metres)	615	182	797	77.16%
MELHEGJ	Electricity (GJ) heating cycle	1,132	3,418	4,550	24.88%
MELCOGJ	Electricity (GJ) air-conditioning cycle	1,141	3,409	4,550	25.08%
MELTOGJ	Electricity (GJ) total of both cycles	1,142	3,408	4,550	25.10%
MNGHEGJ	Natural gas (GJ) heating cycle	902	1,378	2,280	39.56%
MNGCOGJ	Natural gas (GJ) air-conditioning cycle	901	1,379	2,280	39.52%
MNGTOGJ	Natural gas (GJ) total of both cycles	903	1,377	2,280	39.61%
MOILTOGJ	Fuel oil total (GJ)	382	323	705	54.18%
MPROTOGJ	Propane total (GJ)	98	52	150	65.33%

It is recommended that the rate of imputation be taken into account when analysing the variables subject to imputation. In this regard, the reader is referred to in Section 4.3, "Guidelines concerning the quality of estimates."

Energy use

In addition to having undergone high rates of imputation, the energy use variables present another difficulty. Based on a number of verifications made by the project manager, it was noted that consumption data could be of lower quality in the case of duplexes and "apartments in a building with fewer than five storeys." For these dwellings, it seems that the concept "proportion of the building's consumption attributable to the dwelling" was not very successful. Therefore, the consumption data of dwellings resulting from the application of this proportion to the building's consumption data could have been incorrectly estimated. The approach in question had not been tested prior to the survey, which could explain its relative lack of success. In addition, the ambiguity between the terms "dwelling" and "building" in English and between "logement" and "immeuble" in French could have contributed to the inaccuracy of the reported proportions.

4.2 Weighting procedures

The weighting of the SHEU sample involved 11 steps:

1. Adjustment of the CCHS subweight to account for the undercoverage at the dwelling level of the file used to draw the SHEU sample (the CCHS file was missing cases where the household responded but the selected member did not respond)
2. Selection of the SHEU sample
3. Adjustment to account for multiple dwellings
4. Adjustment to account for type-0.1 full non-responses (dwellings whose address was not confirmed)
5. Adjustment to account for type-0.2 full non responses (dwellings whose occupancy was not confirmed
6. Adjustment to account for type-1 full non-responses (no data gathered)
7. Adjustment to account for type-2 full non-responses (households who moved into the selected dwelling in 2004)
8. Adjustment to account for type-1 partial non-responses (data on the heating system and hot water tank, used for determining the energy sources used by the dwelling, were missing)
9. Adjustment to account for type-2 partial non-responses (other cases of partial non-response)
10. Initial calibration of weights to produce estimates consistent with CCHS estimates, with two margins: first, the number of dwellings in scope, by province, and second, the

number of dwellings based on certain groupings of dwelling types and based on a selected geographic level (national or regional)

11. Second calibration of weights to produce estimates consistent with LFS estimates in 10 domains: grouped type of dwelling (single detached dwellings vs. all others) for each of the usual five infranational regions (Atlantic provinces, Quebec, Ontario, Prairie provinces, British Columbia)

4.3 Variance estimation

Use of Bootvar software

The bootstrap method was used for estimating the SHEU variance. Basically, the 500 bootstrap weights of the CCHS were subjected to the 11 steps of the SHEU weighting procedure, as described in Section 4.2. The resulting bootstrap weights were then used for estimating the SHEU variance. The Bootvar software and its documentation accompany the SHEU microdata file so that the variance estimates can be produced.

Coefficients of variation tables

In keeping with the tradition of the previous Surveys of Household Energy Use, coefficients of variation tables have been created for the usual five infranational regions and for the 10 provinces as a whole. The estimates provided in these tables are less precise than those obtained with the bootstrap method. The tables are based on an overall design effect by region corresponding to the third quartile of the design effects calculated using Bootvar for 100 variables representative of the SHEU.

Guidelines concerning the quality of estimates

The coefficient of variation of an estimate is obtained with the Bootvar software. In the case of variables that were not imputed, the guidelines in Table 14 apply.

Table 14
Guidelines concerning the quality of the estimate in the case of non-imputed variables

Level of quality of the estimate	**Guideline**
Acceptable	The estimates are based on a sample of 30 or more units and present low coefficients of variation, between 0.0 and 16.5 percent. Acceptable estimates are marked with the letter A.
Use with caution	The estimates are based on a sample of 30 or more units and present high coefficients of variation, between 16.6 and 33.3 percent. Such estimates are marked with the letter M.
Too unreliable to be published	The estimates are based on a sample with fewer than 30 units, or they present very high coefficients of variation, greater than 33.3 percent. Statistics Canada recommends that such estimates not be released as their quality is unacceptable. Unreliable estimates are marked with the letter U and are accompanied by the following warning: "Estimates marked by the letter U do not meet Statistics Canada's quality standards. Any conclusion based on this data is not dependable and is most likely invalid."

In the case of imputed variables, the procedure to follow for determining the level of quality of the estimate is as follows:

1. Calculate the coefficient of variation (CV) of the estimate with the Bootvar program. Include reported values and imputed values in this estimate and in the CV calculation. A value is deemed reported if its imputation flag equals 3 (not imputed). A value is considered imputed when its imputation flag is 1 (imputed deterministically) or 2 (imputed from donor).

2. Calculate the rate of imputation of the estimate based on the variable's imputation flag: rate of imputation = i/n, where i is the number of imputed values and n is the sum of the number of reported values plus the number of imputed values.

3. Calculate the "adjusted CV" using the CV from Bootvar and the rate of imputation, as follows:

   CV_adjusted = $\sqrt{\frac{n}{r}}\,CV = \frac{CV}{\sqrt{1 - \text{Rate of imputation}}}$

   where r is the number of reported values and n is the sum of the number of reported values plus the number of imputed values. Round the adjusted CV to the nearest tenth of a percent (for example, round 16.58 percent to 16.6 percent).

4. Apply the guidelines described in Table 14 by replacing the term "sample" (which, in the case of imputed variables, equals all reported values plus imputed values, represented here by n) with the term "sample restricted to reported values" (which equals the number of all reported values, represented here by r) and by replacing the expression "coefficient of variation" with "adjusted coefficient of variation" (as defined in Step 3 above).